

negative expected returns. These assets are so attractive because they have large payoffs when the market is crashing. Gold, or sometimes government bonds, are often presented as examples of low (or negative) beta assets which tend to pay off when the stock market crashes. (Government bonds were one of the few asset classes to do well in the financial crisis in 2008. See also chapter 9 on fixed income.)

3.6. CAPM LESSON 6: ASSETS PAYING OFF IN BAD TIMES HAVE LOW RISK PREMIUMS

Another way to view the SML relationship in equation (6.2) is that the risk premium in the CAPM is a reward for how an asset pays off in bad times. Bad times are defined in terms of the factor, which is the market portfolio, so bad times correspond to low (or negative) market returns. If the asset has losses when the market has losses, the asset has a high beta. When the market has gains, the high beta asset also gains in value. Investors are, on average, risk averse so that the gains during good times do not cancel out the losses during bad times. Thus, high beta assets are risky and require high expected returns to be held in equilibrium by investors.

Conversely, if the asset pays off when the market has losses, the asset has a low beta. This asset is attractive and the expected return on the asset can be low—investors do not need much compensation to hold these attractive assets in equilibrium. More generally, if the payoff of an asset tends to be high in bad times, this is a valuable asset to hold and its risk premium is low. If the payoff of an asset tends to be low in bad times, this is a risky asset and its risk premium must be high.

In the CAPM, the bad returns are defined as low returns of the market portfolio. This is, of course, very restrictive: there are many more factors than just the market. In multifactor models, all the intuitions of CAPM Lessons 1 through 6 hold. Except that, with multiple factors, each factor defines its own set of bad times.

4. Multifactor Models

Multifactor models recognize that bad times can be defined more broadly than just bad returns on the market portfolio. The first multifactor model was the *arbitrage pricing theory* (APT), developed by Stephen Ross (1976). It uses the word “arbitrage” because the factors cannot be arbitrated or diversified away—just like the single market factor in the CAPM. In equilibrium, investors must be compensated for bearing these multiple sources of factor risk. While the CAPM captures the notion of bad times solely by means of low returns of the market portfolio, each factor in a multifactor model provides its own definition of bad times.